Todo - inner filter slides

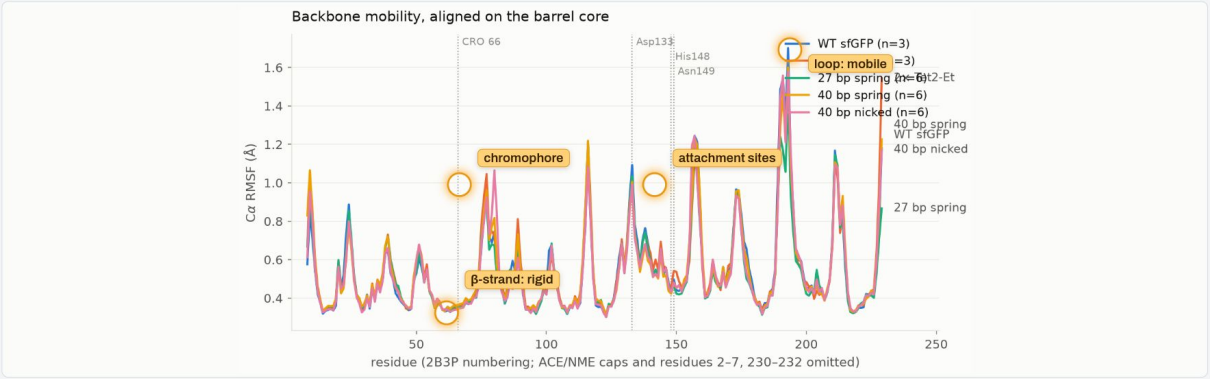
- Explain the purpose of metric is concentration independence
- Inner filter effect only make 16 and above glow
- Change tet quenching attempt 3 concentrations to 1.3,3.3,1.5
- We can assume concentration independence because highest concentration we used for a reading is 10 mu M

Todo -

On the blank slide that has the title(Flurometry) explain that now that we have purified protein sample we seek to see if the addition of tet groups and mechanical tension due to spring tension affects fluroscence. We use a hobira flurometer doing excitation and absorption spectrums(define exactly what eachh of them are) and say that for excitation we sweep from 400 to 500, fix at 485nm. For absorption we sweep from 500 to 600 hold at 510 nm. 485,510 or known emission and excitation wavelengths of sfGFP if im not wrong.

Consider this current slide:

sfGFP: backbone rigidity (RMSF)

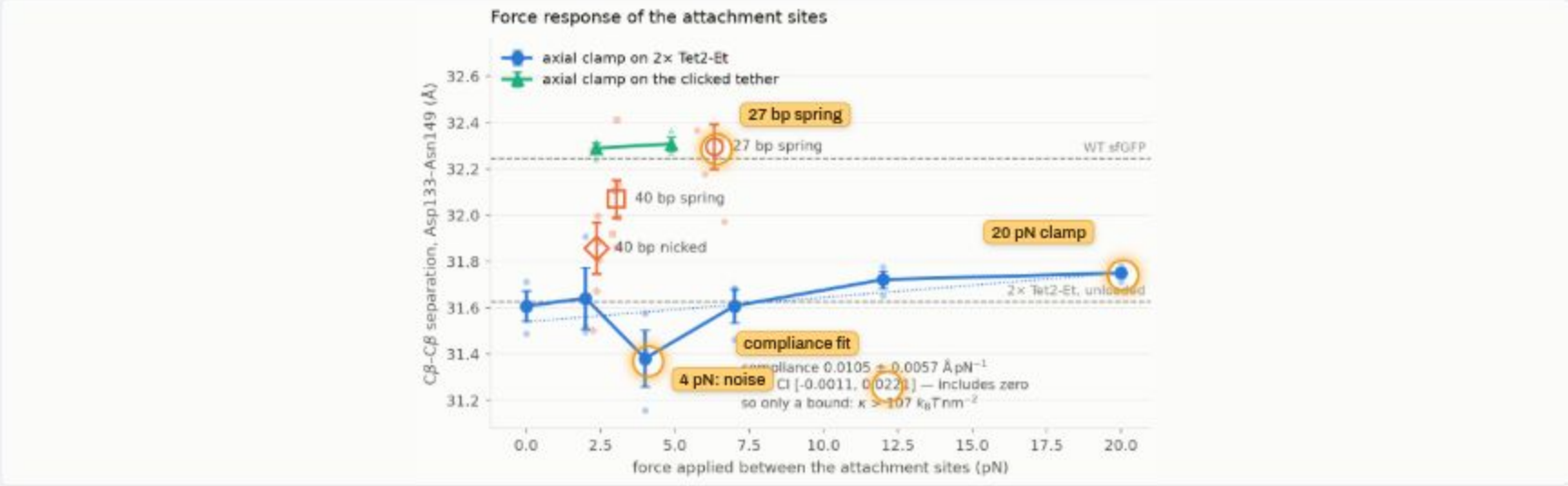


Cα RMSF, aligned on the barrel core

RMSF (root-mean-square fluctuation): how far each residue's Cα atom wanders from its own average position over the trajectory, after every frame is aligned on the rigid core. Low = rigid, high = mobile. x-axis: residue number (2B3P numbering); y-axis: Cα RMSF (Å). Dashed lines mark the chromophore (CRO 66) and the attachment sites Asp133, His148, Asn149. One colour per system: WT, 2× Tet2-Et and the 27 / 40 / 40-nicked bp springs (n = replicates).

Explain what exactly is the loop. It helps if you identify the main structural characteristics of sfGFP, for example label where the beta strand is, or where that “loop is” on the first md analysis slide for sfGGFP. Moreover remove the “50nt ssDNA vs 50nt dsDNA closed loop” in the systems compared. Just say various lengths of DNA springs, ds vs ss, clicked vs unclicked etc. because we clearly included lengths under then 50nt in our MD plots.

sfGFP: backbone rigidity (RMSF)



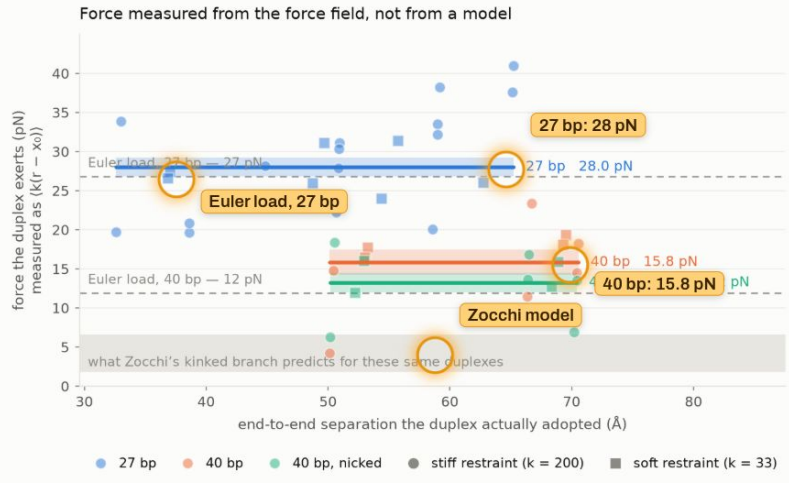
Axial force clamp between the two Cβ atoms

Axial force clamp: a constant force applied directly between the two Cβ atoms, a controlled stand-in for a spring. x-axis: applied force (pN); y-axis: Cβ–Cβ distance Asp133–Asn149 (Å). **Blue**: clamp on the unclicked protein; **green**: clamp on the clicked tether; **orange**: the DNA-spring chimeras at their modelled force. **Compliance** is the slope (Å per pN); the **stiffness κ** is its inverse.

This slide is too confusing. I want you to just show the blue plot only. Ignore the rest . the main point here is just to demonstrate even without force application the backbone is already super rigid

Again this slide is also very confusion. What is k what is r what is x0 what is end to end separation what is duplex. What are the random shapes all over the plot. What do we learn from this. I suggest your render the latex equations explicitly and explain the terms

sfGFP: forces applied, force field vs model

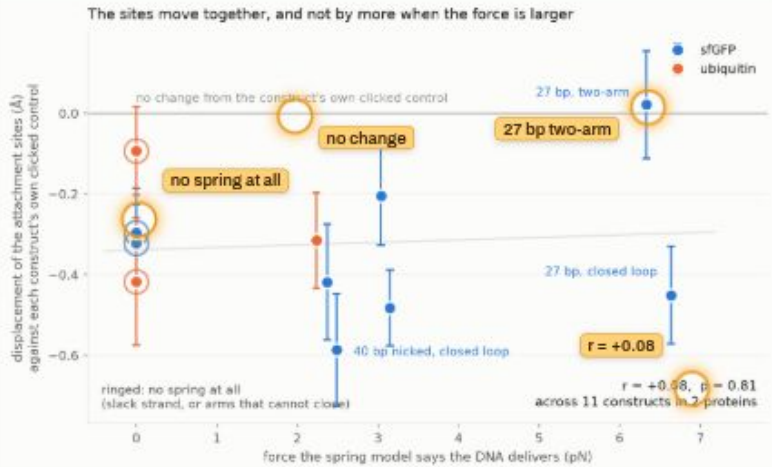


Force measured on the bare duplex, restraint windows

A bare duplex, no protein, held at a series of end-to-end separations by a harmonic restraint between its two linker nitrogens. x-axis: the separation the duplex actually adopted (Å); y-axis: the force the duplex exerts on the restraint, $\langle k(r - x_0) \rangle$ (pN). Points are runs, lines are means. Dashed: the **Euler buckling load** $\pi^2 B/L^2$, the force at which a straight elastic rod buckles. Grey band: what **Zocchi's kinked-rod model** predicts for the same duplexes.

Just remove this plot from the slides altogether. In even contains ubiquitin which is a system we are not even interested in

sfGFP: forces applied, force field vs model

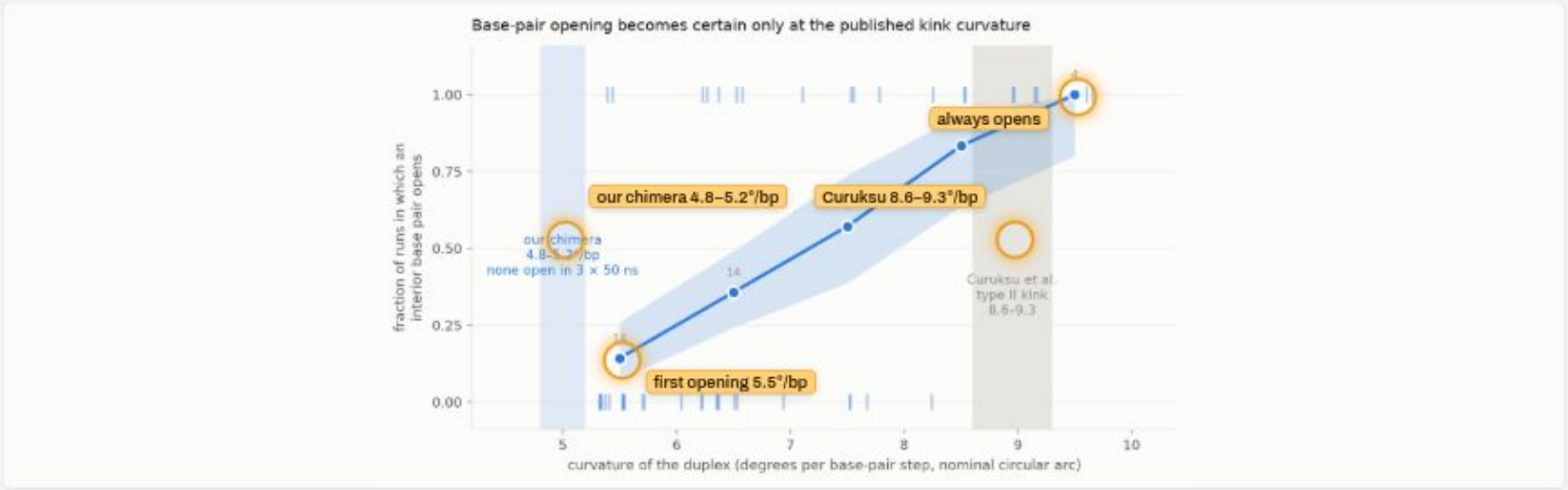


Site displacement against modelled force

x-axis: the force the spring model says each DNA delivers (pN). y-axis: the shift of the attachment sites against each construct's own clicked control (Å); below zero = pulled together. **Blue** sfGFP, **orange** ubiquitin (preprint). Ringed points carry no spring at all (slack strand or arms that cannot close). A real force effect would climb towards the upper right.

Just remove this plot altogether as well. I am not interested in base pair opening related graphs as well

sfGFP: DNA dynamics

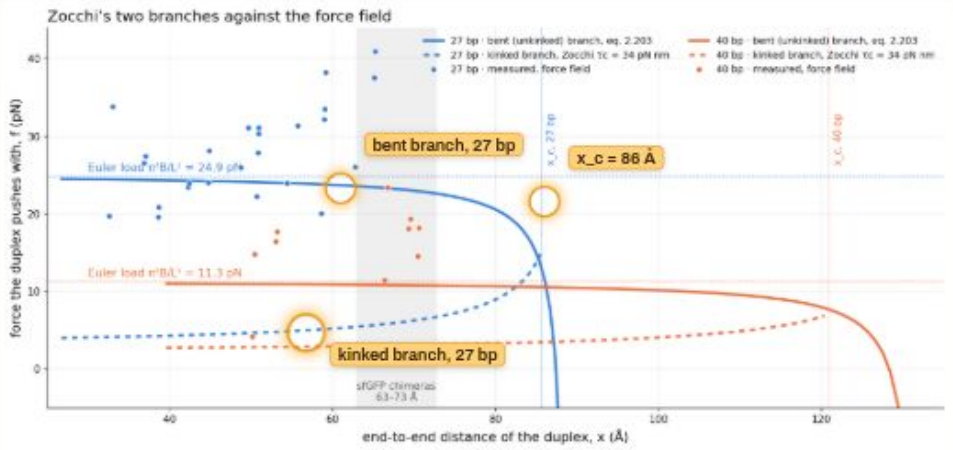


Base-pair opening against imposed curvature

Bare 27 and 40 bp duplexes held at set curvatures. x-axis: curvature (° per base-pair step); y-axis: fraction of runs in which an **interior** base pair opens. Ticks are single runs (top = opened, bottom = stayed paired). **Blue band**: where our chimera sits; **grey band**: the type II kink range Curuksu et al. found by umbrella sampling.

Remove the plot but keep the analysis, instead i want you to render the equations latex explicitly and then explain the terms carefully. Do not use the word bent branch. Use kink or unkinked. Besides you did not mathematically show unkined branch is equivalent to the force field measurements mathematically like i requested you to at all

sfGFP: why the force field disagrees with Zocchi's model



Zocchi's two force branches (lines, τ_c = 34 pN nm, B = 200 pN nm²) and the force field's bare-duplex measurements (points)

Zocchi's DNA spring (*Molecular Machines* §2.15). A duplex of contour $L = 0.33\ \text{nm} \times N_{bp}$ and bending modulus $B = k_B T \cdot \ell_p \approx 200$ pN nm² (persistence length $\ell_p \approx 50$ nm), held at end-to-end distance x, pushes its ends apart with:

bent branch ($x > x_c$): $f = 10B/L^2 - k_BT/(L - x)$ (eq. 2.203)

kinked branch ($x < x_c$): $f = \tau_c / [2R\sqrt{(1 - (x/2R)^2)}]$, the slope of $E_k = \tau_c \arccos(x/2R)$ (eq. 2.196), $R = (L/2)(1 - \gamma^2/90)$

The rod switches at $x_c = L(1 - \gamma^2/15)$, $\gamma = L\tau_c/2B$ (eqs 2.188, 2.194), where its bending torque reaches the **critical torque** $\tau_c \approx 27$ (nicked) to 34 pN nm (continuous).